UNIT 2 · TOPIC 8(B)

Thresher Design

Syllabus topic: Design of harvesters and threshers

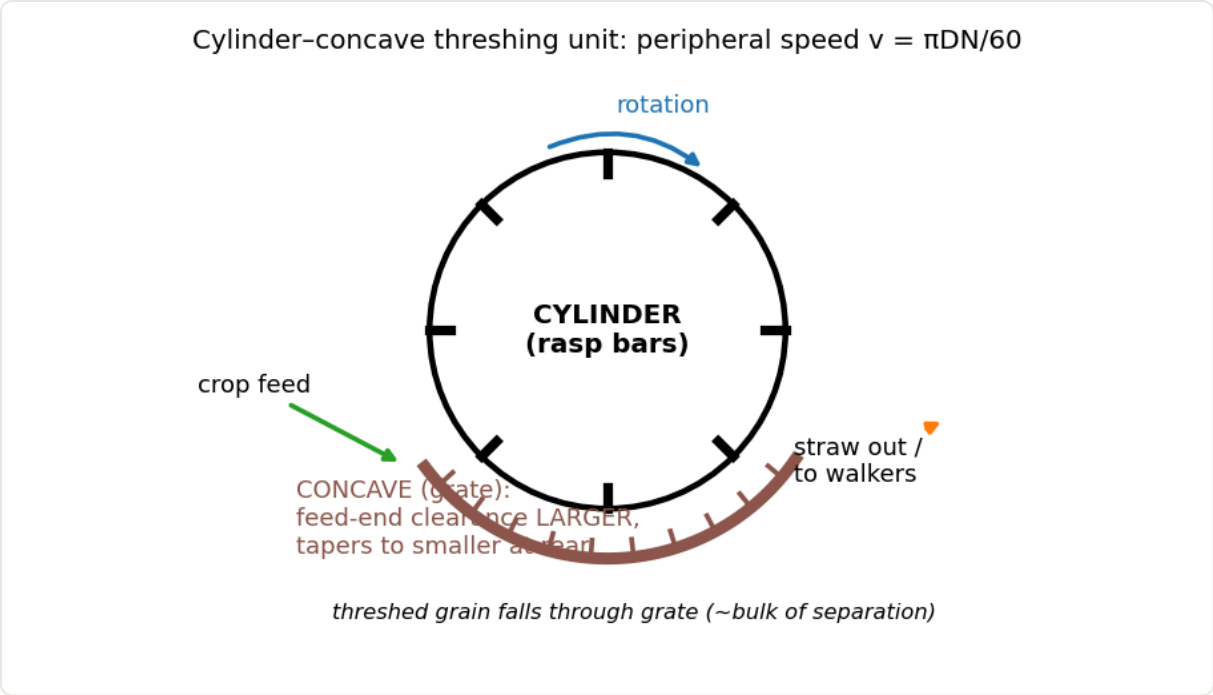
Threshing principle

Threshing detaches grain from the ear/panicle/pod mainly by **impact** of fast-moving cylinder elements, plus **rubbing** (attrition) between cylinder, crop mat and concave, and some combing. Its effectiveness depends on:

- Cylinder peripheral speed.
- Cylinder–concave clearance.
- Type of cylinder.
- Feed rate and crop condition (moisture!).

Cylinder types

Cylinder	Action / use
Rasp-bar	Corrugated bars; impact + rubbing as the crop is dragged through the concave gap; standard on combines and multicrop use; wheat peripheral speed 25–30 m/s (5000–6000 fpm).
Spike-tooth (peg-tooth)	Rows of spikes pass between concave teeth — aggressive combing; common in Indian wheat threshers; more straw breakup (bhusa making).
Angle-bar	Rubber-faced angle bars; gentler; once used for delicate crops/clover.
Wire-loop	Loops of wire on a drum; "hold-on" paddy threshers (pedal/power) — heads combed while straw is held; ~12–15 m/s keeps paddy undamaged.
Hammer-beater/"drummy" and syndicator	Beat and break the whole crop; fine bhusa (wheat, N-India).
Axial-flow	The crop spirals around the cylinder axis several times (multiple gentle impacts) → high threshing efficiency, less damage, but more straw breakup and power; standard for paddy (axial-flow paddy thresher) and modern rotary combines.
Rubber-roll	Two rubber rolls at 250–300 fpm (1.3–1.5 m/s) press the pods — for easily damaged podded seeds (beans); minimal seed damage.



Cylinder–concave threshing unit

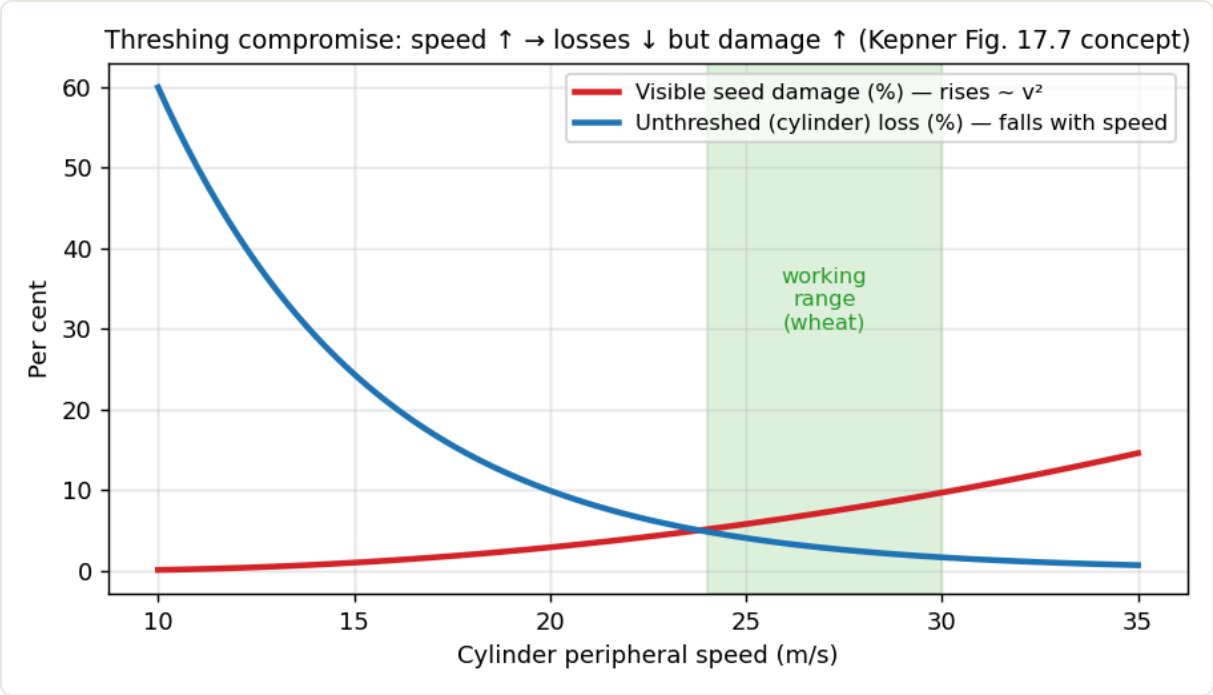
Design parameters

$$v = \pi DN/60 \text{ (m/s)}; N = 60 \cdot v/(\pi D)$$

Choose v for the crop first, then compute the rpm from the available cylinder diameter.

Ex.1 Cylinder rpm. Wheat needs v = 28 m/s on a 55-cm cylinder: N = 60 × 28/(π × 0.55) ≈ **972 rpm**.

- **Cylinder–concave clearance:** larger at the entry (feed) side, tapering smaller toward the rear — the crop mat is compressed progressively. Closer clearance and higher speed → more complete threshing but more seed damage, straw breakup and power draw.
- **Concave:** a curved grate of bars/wires under **90–120°** of the cylinder; most of the threshed grain separates through the concave-and-grate openings immediately.
- **Speed–damage–loss compromise:** unthreshed (cylinder) loss **falls** steeply as peripheral speed rises, while seed damage **rises** roughly with the square of speed. Choose the lowest speed that gives acceptable threshing; open the clearance before reducing speed for damage-sensitive seed. Germination damage matters even when no damage is visible (internal cracks).



Threshing compromise — speed vs losses and damage

- **Feed rate:** capacity (kg/h of crop) rises with feed rate until threshing/separating losses climb; thresher tests (IS/RNAM codes — Unit 3) report capacity at specified loss limits.

Performance definitions

$$\text{Threshing efficiency} = (\text{grain threshed} \div \text{total grain input}) \times 100 = 100 - \%\ \text{unthreshed grain}$$

$$\text{Cleaning efficiency} = \text{clean grain} \div (\text{grain} + \text{chaff in grain outlet}) \times 100$$

- Target threshing efficiency **≥ 98–99%**.
- Total losses = unthreshed + broken/damaged + blown + spilled + separation loss.
- Power requirement rises with feed rate and moisture; express specific energy in kWh per tonne of crop or grain. A blower/sieve unit adds cleaning (aspirator-type vs screen-cleaner threshers).

Ex.3 Threshing efficiency. Input grain 500 kg/h; unthreshed in straw + tailings 5 kg/h → efficiency = (495/500) × 100 = **99%**.

Safety

Feeding-chute geometry per **IS 9129** (safe feeding length/height so the operator's hand cannot reach the cylinder) — thresher injuries are a major Indian farm-accident category (Unit 5).

Note: BIS has withdrawn IS 9129 (1979, safe feeding systems for power threshers); IS 9020 — Power Threshers: Safety Requirements (revised 2002) now covers it. Exam keys may cite either standard.

Practice MCQs

Q155. The wire-loop cylinder is characteristic of:

(a) Maize sheller
(b) 'Hold-on' paddy threshers
(c) Groundnut decorticator
(d) Sunflower thresher

Q156. A rasp-bar cylinder threshes mainly by:

(a) Pure shear
(b) Impact plus rubbing of crop between bars and concave
(c) Centrifuging
(d) Suction

Q157. Cylinder peripheral speed is given by:

(a) $\pi DN/60$
(b) $60/\pi DN$
(c) $\pi D/60N$
(d) $DN/60$

Q158. For wheat threshing at 28 m/s with a 55-cm cylinder, required rpm ≈

(a) 490
(b) 720
(c) 972
(d) 1250

Q159. Raising cylinder speed at fixed clearance generally:

(a) Reduces threshing and damage
(b) Increases threshing completeness AND seed damage
(c) Increases losses only
(d) Has no effect

Q160. Cylinder–concave clearance is normally:

(a) Uniform
(b) Larger at feed end, tapering smaller at rear
(c) Smaller at feed end
(d) Zero at entry

Q161. Visible seed damage rises approximately with:

(a) $\sqrt{v}$
(b) v
(c) v^2
(d) v^3 of cylinder peripheral speed

Q162. For threshing easily damaged podded seeds (e.g., beans), the gentlest unit is:

(a) Spike-tooth cylinder
(b) Rubber rolls at 1.3–1.5 m/s
(c) Hammer beater
(d) High-speed rasp bar

Q163. Threshing efficiency with 2% unthreshed grain is:

(a) 2%
(b) 96%
(c) 98%
(d) 100%

Q164. The axial-flow thresher achieves high efficiency with low damage because:

(a) Very high speed
(b) Crop receives many gentler impacts while spiralling around the cylinder
(c) No concave
(d) Larger seeds

Q165. Indian standard governing thresher feeding-chute safety is:

(a) IS 4468
(b) IS 9129
(c) IS 6284
(d) IS 2062

Q166. In Indian wheat threshers, fine bhusa is produced chiefly by:

(a) Rubber rolls
(b) Beater/'drummy' and spike-tooth types
(c) Wire loop
(d) Stripper header

Answers

Q155 — (b) 'Hold-on' paddy threshers

Q156 — (b) Impact plus rubbing of crop between bars and concave

Q157 — (a) $\pi DN/60$

Q158 — (c) 972 — *$N = 60v/\pi D = 60\times28/(3.1416\times0.55) = 972$ rpm.*

Q159 — (b) Increases threshing completeness AND seed damage

Q160 — (b) Larger at feed end, tapering smaller at rear

Q161 — (c) v^2 — *Kinetic energy of impact $\propto v^2$ — Kepner Fig. 17.7 shows damage climbing steeply while unthreshed loss falls as speed rises; the design compromise defines the working speed band.*

Q162 — (b) Rubber rolls at 1.3–1.5 m/s

Q163 — (c) 98% — *Threshing efficiency = 100 - % unthreshed = 98%.*

Q164 — (b) Crop receives many gentler impacts while spiralling around the cylinder

Q165 — (b) IS 9129

Q166 — (b) Beater/'drummy' and spike-tooth types

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 14 (Hay Harvesting: Mowing) and Ch. 17 (Grain and Seed Harvesting); Jagdishwar Sahay — Elements of Agricultural Engineering; BIS thresher standards (IS 9129 safety); NPTEL Farm Machinery lectures.